

13. Loops in java

Looping Constructs in Java – Theoretical Overview

Java provides several types of loops to perform repetitive tasks efficiently. Each loop serves a different purpose depending on the nature and requirement of the iteration.

1. For Loop

Definition:

The `for` loop is an **entry-controlled loop** used when the number of iterations is known beforehand.

Syntax-

```
1 for (initialization; condition ; increment/decrement) {  
2     // code block to be executed  
3 }
```

Explanation:

- **Initialization:** Executed once at the start of the loop.
- **Condition:** Evaluated before each iteration. If `true`, the loop body is executed.
- **Update:** Executed after each iteration to modify the loop variable.

Example:

```
1 for (int i = 0; i < 5; i++) {  
2     System.out.println("Count: " + i);  
3 }
```

1.1 nested-for loop

Syntax -

```
1 // Outer loop  
2 for (int i = 1; i <= 2; i++) {  
3     System.out.println("Outer: " + i); // Executes 2 times  
4  
5     // Inner loop  
6     for (int j = 1; j <= 3; j++) {  
7         System.out.println(" Inner: " + j); // Executes 6 times (2 * 3)  
8     }  
9 }  
10
```

2. While Loop

Definition:

The **while** loop is an **entry-controlled loop** used when the number of iterations is not known in advance. It checks the condition before executing the loop body.

Syntax -

```
1 while (condition) {  
2     // code block to be executed  
3 }
```

Explanation:

- The condition is checked before the loop body is executed.
- If the condition is **true**, the loop body is executed.
- If the condition becomes **false**, the loop terminates.

example -

```
1 int i = 0;  
2 while (i < 5) {  
3     System.out.println(i);  
4     i++;  
5 }
```

3. Do-While Loop

Definition:

The do-while loop is an exit-controlled loop, meaning the loop body is executed at least once before the condition is checked.

Syntax -

```
1 do {  
2     // code block to be executed  
3 }  
4 while (condition);
```

Explanation:

- Executes the loop body once, then checks the condition.
- If the condition is **true**, it repeats the loop.
- Guarantees **at least one execution**.

example

```
1 int i = 0;  
2 do {  
3     System.out.println(i);
```

```
4     i++;  
5 }  
6 while (i < 5);
```

4. For-Each Loop

Definition:

The for-each loop (also called enhanced for loop) is used to iterate over arrays or collections without needing an index variable.

Syntax -

```
1 for (type variableName : arrayName) {  
2     // code block to be executed  
3 }
```

Explanation:

- Simplifies the traversal of arrays and collections.
- Automatically fetches each element in order.
- Does **not allow modification** of the underlying collection during iteration.

example

```
1 String[] cars = {"Volvo", "BMW", "Ford", "Mazda"};  
2 for (String i : cars) {  
3     System.out.println(i);  
4 }
```

“Programming isn’t about what you know; it’s about what you can figure out.” - Chris Pine

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